

ECG Anomaly Detection — Session Report

Prepared for: Verification • Session ID: 690c7b96 • Generated: 2026-08-24 07:31:51

Session Information

Patient / User	Sriyantha Sadaruwan
Source	mit-bih-arrhythmia (100)
Started	2026-08-24 07:31:51
Ended	2026-08-24 07:32:20
Duration	29.1 s
Status	completed
Total beats analyzed	0
Anomalous beats	0 (0.00%)
Unique alerted beats in report	0
Model	MIT-BIH Normal-Sinus Autoencoder (nsrdb-primary-20260802) • Dilated-U-Net-Autoencoder
Analysis mode	Beat-aligned hierarchical morphology + RR
Abnormal-probability threshold	0.6208

Research Validation Context

This section reports model-validation evidence separately from the patient/session recording above. It must not be read as a diagnosis for this session.

Final Report benchmark	Value
Protocol	Final within-patient temporal holdout
Generalization claim	unseen temporal signal from known patients; not unseen-patient generalization
Binary accuracy	98.47%
Binary balanced accuracy	98.22%
Whole-system accuracy	96.86%
Whole-system macro F1	0.9303

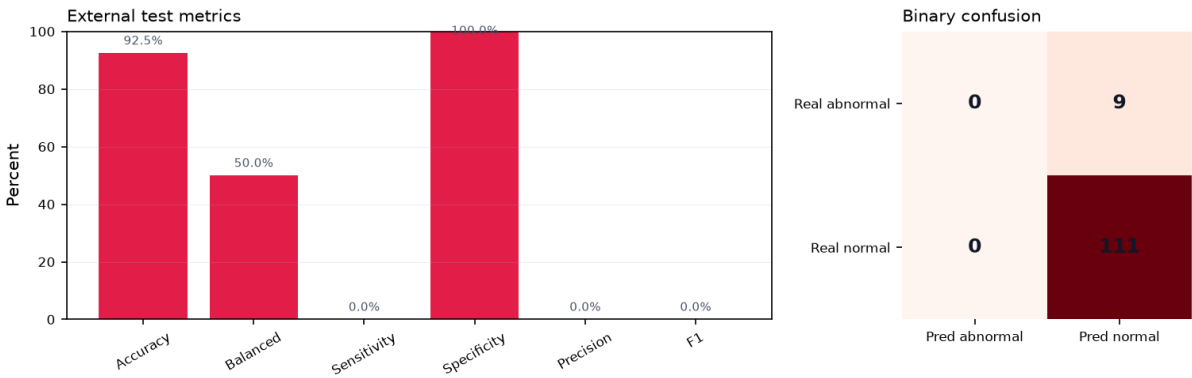
The Final Report states that the fixed MITDB result is later signal from known patients and is not proof of cold-start performance on completely unseen patients.

Totally unseen external dataset test Latest saved result	
Dataset	St Petersburg INCART 12-lead Arrhythmia Database
Lead / mode	II / offline
Saved at	2026-08-24T09:36:17.318905+00:00
Records scored	1
Annotated beats	120
Threshold	0.1589
Accuracy	92.50%
Balanced accuracy	50.00%
Sensitivity / specificity	0.00% / 100.00%
F1 / precision	0.00% / 0.00%
AUROC / AUPRC	0.3293 / 0.0585
Confusion TP/TN/FP/FN	0 / 111 / 0 / 9

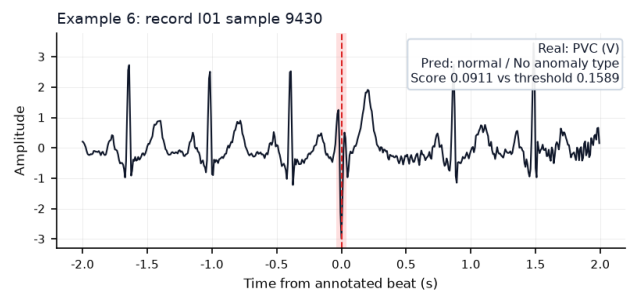
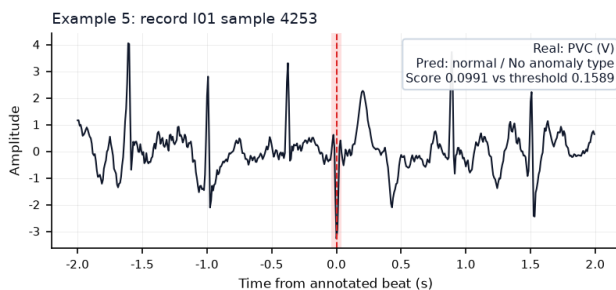
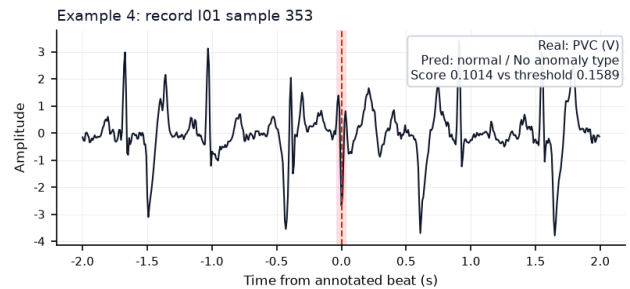
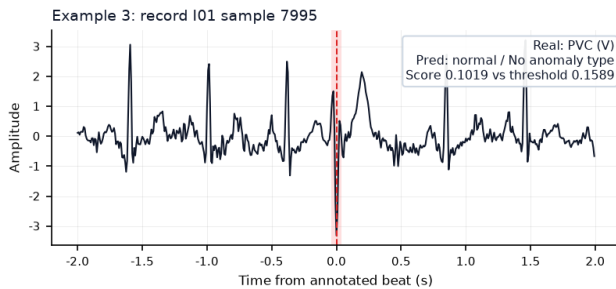
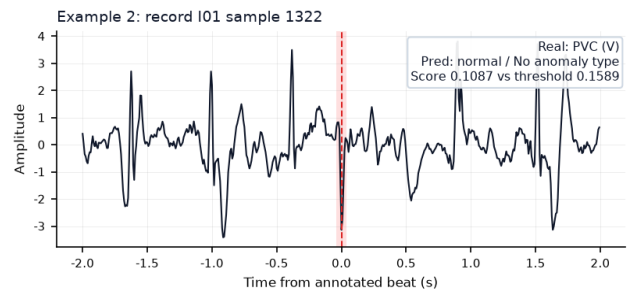
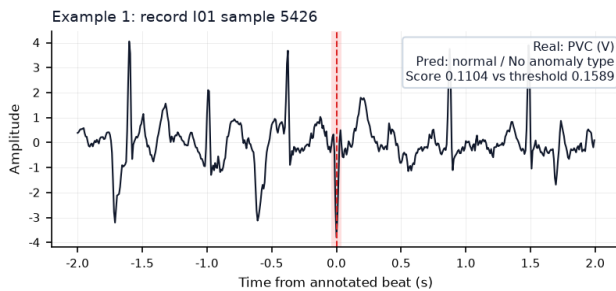
External validation: INCART lead II is not identical to MIT-BIH MLII, so results measure dataset, device, and lead-domain shift.

External Test Figures

The chart summarizes the saved external INCART evaluation. The example segments below are centered on annotated beats and show real annotation type, predicted anomaly decision, predicted anomaly type when available, and the score-threshold comparison.



Example	Record / sample	Real type	Predicted decision	Predicted type	Score
1	I01 / 5426	PVC (V)	Normal	No anomaly type	0.1104 vs 0.1589
2	I01 / 1322	PVC (V)	Normal	No anomaly type	0.1087 vs 0.1589
3	I01 / 7995	PVC (V)	Normal	No anomaly type	0.1019 vs 0.1589
4	I01 / 353	PVC (V)	Normal	No anomaly type	0.1014 vs 0.1589
5	I01 / 4253	PVC (V)	Normal	No anomaly type	0.0991 vs 0.1589
6	I01 / 9430	PVC (V)	Normal	No anomaly type	0.0911 vs 0.1589



ECG Recording & Detection Results

The following plot shows the recorded ECG signal (z-score normalized). Red markers indicate R-peak-aligned beats whose supervised abnormal probability exceeded the validation-selected threshold.

No ECG data recorded for this session.

Main Arrhythmia Finding

No alerts were triggered during this session. No analysed beat exceeded the abnormal-probability threshold.

Clinical Summary

Fewer than 1% of analysed beats exceeded the model threshold. This research result does not exclude arrhythmia.

Physician Notes

Report verification with external INCART figures.

Physician signature: _____

Date: _____

This report is generated automatically by a research ECG anomaly-detection system and is intended for research review only. Model probability and subtype confidence are not clinical certainty. This report does not constitute a medical diagnosis and must not replace qualified clinical judgment.